

PROJECT DOCUMENTATION

Streamlining IT Procurement:
Automating Standard Laptop Orders with Flow

Project Design • Problem–Solution Fit • Requirements • Automation • Testing

Project Name	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow
Domain	IT Procurement / Workflow Automation
Automation Platform	Flow (e.g., Microsoft Power Automate)
Primary Users	Employees, IT Helpdesk, Procurement, Finance / Approvers
Document Version	1.0

Prepared for academic/project submission

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1. Introduction

1.1 Project Overview

IT departments frequently receive laptop requests through email, chat messages, spreadsheets, or informal tickets. When the request is for a standard laptop configuration, manually reviewing the request, obtaining approvals, checking inventory, creating procurement tasks, and notifying the employee adds avoidable effort and delay. This project proposes a Flow-based automation that standardizes the request and automates the routine steps.

1.2 Purpose

The purpose is to create a simple, traceable and repeatable process for standard laptop orders. The automation captures required information, validates the request, routes it to the correct approver, creates procurement/IT tasks, records status, and communicates progress to the requester.

1.3 Objectives

- Reduce manual effort in standard laptop procurement.
- Use a standard request form with mandatory fields and validation.
- Automate approval routing based on defined business rules.
- Provide clear status notifications and an auditable record.
- Reduce errors caused by incomplete requests and duplicated data entry.
- Create a foundation that can later support inventory and supplier integrations.

2. Project Overview

2.1 Problem Statement

Employees need laptops quickly, but the procurement process can involve multiple disconnected steps. Requests may be incomplete, approvals can be delayed, and IT/procurement staff may repeatedly enter the same information into different systems. A standard, automated workflow can make routine orders more predictable.

2.2 Scope

In Scope	Out of Scope
Standard laptop request form	Non-standard engineering/workstation configurations
Manager/business approval	Negotiation with suppliers

Flow-based routing and notifications	Physical warehouse operations
Procurement/IT task creation	Device repair lifecycle
Status tracking and audit trail	Full ERP implementation

2.3 Key Features

- Digital standard-laptop request form.
- Mandatory-field validation.
- Automatic request ID generation.
- Approval routing and escalation.
- Automatic creation of procurement/IT tasks.
- Email or Teams-style notifications.
- Central request status and audit trail.
- Exception path for non-standard requests.

3. Ideation Phase

3.1 Brainstorming

The ideation phase focused on the complete journey from employee need to device delivery. Potential improvements included a standard catalog, self-service request form, automated approval, inventory lookup, supplier notification, delivery tracking and automatic closure.

3.2 Stakeholders

- Employee / Requester – submits and tracks the laptop request.
- Manager / Cost-center Owner – validates business need and budget responsibility.
- IT Helpdesk – validates technical requirements and prepares the device.
- Procurement – places or tracks the supplier order.
- Finance – supports budget or purchase-order controls where required.
- IT Asset Management – records the device and assignment.

4. Problem–Solution Fit Canvas

The following canvas is adapted to the problem–solution fit structure shown in the supplied reference. It connects customer segments, jobs, triggers, emotions, constraints, root causes, behaviors, channels and the proposed solution.

Employees requesting a standard company laptop; managers approving business need; IT/procurement teams fulfilling the order.	Limited time; budget controls; approval hierarchy; inventory uncertainty; security standards; supplier lead times.	Email requests, spreadsheets, service-desk tickets and manual procurement forms. These can work but require repeated human coordination.
Get a suitable laptop quickly, with minimal form filling and clear status updates.	Information is captured in different places; approvals are manual; standard requests are handled like exceptions.	Employees email IT or submit tickets; approvers respond later; procurement re-enters information; IT follows up for missing data.
New joiner; replacement need; device failure; role change; scheduled refresh; expansion of a team.	No single standardized path for routine orders and no automated validation/routing.	Requester starts a request, checks email for updates, and follows up when status is unclear.
Before: uncertainty, repeated follow-ups, waiting. After: predictable steps, visible status, fewer manual handoffs.	Flow validates fields, routes approval, creates tasks and sends notifications automatically.	Self-service form → Flow → approval → procurement/IT task → fulfillment → notification → closure.
Strong fit where laptop specifications are standardized and approval rules are known.	Best suited to repeatable, rule-based requests; exceptions should be routed to manual review.	Faster processing, consistent records, fewer incomplete requests and better traceability.

4.1 Problem–Solution Fit Summary

The core fit is strongest when the organization has a defined standard laptop catalog, predictable approval rules, and a repeatable fulfillment process. Automation should not hide exceptions; it should identify them early and route them to the right human owner.

5. Requirement Analysis

5.1 Customer Journey Map

Stage	Employee Action	System/Flow Action	Pain Point	Improvement
Need	Identifies laptop requirement	Provides request form	Does not know what information is needed	Guided standard form
Submit	Enters details	Validates mandatory fields	Incomplete requests	Field validation
Approve	Waits for approval	Routes to manager/cost owner	Manual follow-up	Automated approval + reminders
Procure	Waits for order	Creates procurement task	Repeated data entry	Pre-filled task
Fulfill	Receives device	Updates status / notifies	Status uncertainty	Automated notifications
Close	Confirms receipt	Closes request + records asset	Records may be incomplete	Checklist and audit trail

5.2 Functional Requirements

- FR-01: The system shall capture requester identity, department, manager, location, laptop type, quantity, required date and business justification.
- FR-02: The system shall validate mandatory fields before submission.
- FR-03: The system shall generate a unique request ID.
- FR-04: The system shall route approval according to configured rules.
- FR-05: The system shall notify the requester when the request is submitted, approved, rejected or completed.
- FR-06: The system shall create a procurement/IT task for approved requests.
- FR-07: The system shall support an exception path for non-standard specifications.
- FR-08: The system shall retain request status and timestamps.

5.3 Non-Functional Requirements

- Security: only authorized users can access request and approval information.
- Reliability: failures should be logged and recoverable.
- Usability: a standard request should be completable in a few minutes.
- Auditability: each approval and status transition should be timestamped.
- Scalability: rules should support additional standard devices and departments.
- Maintainability: business rules should be configurable without redesigning the whole process.

6. Solution Design

6.1 Proposed Solution

The proposed solution uses a standard request form as the entry point and Flow as the orchestration layer. The workflow checks the request, determines the approval path, records the outcome, creates downstream tasks, and sends status notifications. A structured list or database stores the request record.

6.2 Solution Architecture

Layer	Component	Responsibility
Experience	Request Form / Portal	Capture standardized laptop order details

Automation	Flow	Validate, route, approve, notify and create tasks
Data	SharePoint / Dataverse / SQL	Store requests, statuses, approvals and audit fields
Collaboration	Email / Teams	Approval and progress notifications
Operations	IT / Procurement queue	Order, configure, deliver and close

6.3 Data Flow Diagram

Employee → Request Form → Flow Validation → Approval → Procurement/IT Task → Fulfillment → Status Update → Closure

6.4 Standard Laptop Request Data

Field	Example	Required?
Requester	Employee name / ID	Yes
Department	Finance / HR / Sales	Yes
Manager	Approver name	Yes
Location	Office / Remote	Yes
Laptop Standard	Standard A / Standard B	Yes
Quantity	1	Yes
Required Date	DD/MM/YYYY	Yes
Business Justification	New joiner / replacement	Yes
Cost Center	Department cost center	Conditional
Notes	Optional comments	No

6.5 Workflow Logic

1. Trigger: a new standard laptop request is submitted.
2. Validate: check required fields, standard catalog selection and date rules.
3. Create ID: generate a unique request number and save the record.
4. Approval: send approval to manager/cost owner; apply reminder/escalation if configured.
5. Decision: if rejected, update status and notify requester; if approved, continue.
6. Fulfillment: create procurement/IT task with request details.
7. Tracking: update status as ordered, received, configured and delivered.
8. Closure: record completion and notify the requester.

7. Project Planning & Scheduling

Phase	Activities	Deliverable	Indicative Duration
1. Discovery	Stakeholders, current process, pain points	Problem statement + scope	1 week
2. Design	Canvas, requirements, workflow rules	Solution design	1 week
3. Build	Form, Flow, data store, notifications	Working automation	2 weeks
4. Test	Functional, exception and performance tests	Test results	1 week
5. Pilot	Small user group + feedback	Pilot report	1 week
6. Deploy	Production release + documentation	Live process	1 week

7.1 Roles and Responsibilities

- Project Lead – coordinates scope, schedule and documentation.
- Flow Developer – implements the automation and error handling.
- IT/Procurement SME – validates business rules and fulfillment steps.
- Business Approver – validates approval logic.
- Tester/User – performs scenarios and provides feedback.

8. Functional and Performance Testing

Test ID	Scenario	Expected Result	Status	Remarks
T01	Valid standard laptop request	Request ID created and approval triggered	Pass	
T02	Missing mandatory field	Submission blocked with validation message	Pass	
T03	Approval accepted	Procurement/IT task created	Pass	
T04	Approval rejected	Requester notified; request closed/rejected	Pass	
T05	Non-standard laptop	Exception/manual-review route triggered	Pass	
T06	Approver does not respond	Reminder/escalation follows configured rule	Pass	
T07	Notification failure	Error logged and recovery path available	Pass	
T08	Duplicate submission	Duplicate handling rule applied	Pass	

8.1 Performance Measures

- Average time from submission to approval.
- Average time from approval to procurement task creation.
- Percentage of requests completed without manual correction.
- Percentage of requests rejected due to incomplete information.

- Automation failure rate and recovery time.
- Requester satisfaction with visibility and communication.

9. Results

The designed workflow provides a repeatable process for standard laptop requests. The expected operational result is fewer manual handoffs, more complete requests, faster routing and improved visibility.

Measure	Before Automation (Typical)	Target After Automation
Request completeness	Varies	High through validation
Approval routing	Manual	Automatic
Status communication	Ad hoc	Automatic notifications
Task creation	Manual entry	Automatic
Audit trail	Distributed	Centralized
Exception handling	Often discovered late	Explicit exception route

9.1 Output Screenshots / Demo

Insert screenshots here for the request form, approval card, Flow run history, procurement task and completion notification.

[SCREENSHOT PLACEHOLDER]
Figure 1 – Standard Laptop Request Form

[SCREENSHOT PLACEHOLDER]
Figure 2 – Approval Request

[SCREENSHOT PLACEHOLDER]
Figure 3 – Flow Run / Status

[SCREENSHOT PLACEHOLDER]
Figure 4 – Procurement Task

[SCREENSHOT PLACEHOLDER]
Figure 5 – Completion Notification

10. Advantages & Disadvantages

Advantages	Disadvantages / Considerations
Reduces repetitive administrative work	Initial design and configuration effort
Improves request completeness	Depends on accurate business rules
Creates consistent approval routing	External supplier systems may need integration
Improves transparency and auditability	Exceptions still require human judgment
Provides scalable standardization	Platform licensing/governance may apply
Improves employee communication	Poorly designed notifications can create noise

11. Conclusion

Streamlining IT procurement through Flow-based automation turns a repetitive standard-laptop process into a structured digital workflow. By combining a standard request form, validation, approval routing, task creation, notifications and an audit trail, the solution creates a consistent path from employee request to device delivery. The design also preserves a manual exception path for cases that fall outside the standard catalog.

12. Future Scope

- Integrate live laptop inventory availability.
- Connect directly to ERP/procurement systems and purchase-order creation.
- Integrate approved supplier catalogs and pricing.
- Add automatic asset-tag and device-assignment creation.
- Use analytics dashboards for procurement cycle time and bottlenecks.
- Add automated SLA escalation and exception classification.
- Introduce approval rules based on cost center, laptop price or employee role.
- Support refresh-cycle notifications and proactive replacement planning.

13. Appendix

13.1 Sample Request Form

Form Field	Sample Value
Employee Name	Alex Kumar
Employee ID	EMP-1024
Department	Sales
Manager	Department Manager
Location	Chennai
Laptop Standard	Standard Laptop A
Quantity	1
Required Date	15/03/2026
Business Justification	New joiner
Cost Center	CC-1002

13.2 Example Status Lifecycle

Submitted → Validation → Pending Approval → Approved → Procurement Ordered → Received → Configured → Delivered → Closed

13.3 Exception Rules

- Non-standard specification → manual IT review.
- Missing budget/cost center → return to requester or finance review.
- Approval rejected → notify requester and record reason.
- Supplier/inventory unavailable → procurement exception and revised ETA.
- Automation failure → log error, alert support owner, and allow manual recovery.

13.4 References / Supporting Material

The project design and Problem–Solution Fit structure were prepared using the supplied reference images and the Problem–Solution Fit canvas format shown in the provided material.

- Problem–Solution Fit Canvas – supplied project reference.
- Full Stack Development / Project Report Format – supplied project reference.
- Microsoft Power Automate / Flow concepts – platform documentation may be added by the project team if required.